VIT-AP UNIVERSITY

Fundamentals of Electrical and Electronics Engineering (FEEE)

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Lecture-36

Faculty

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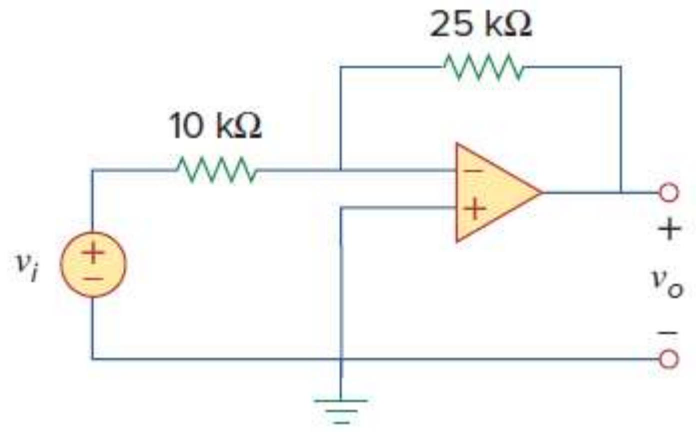
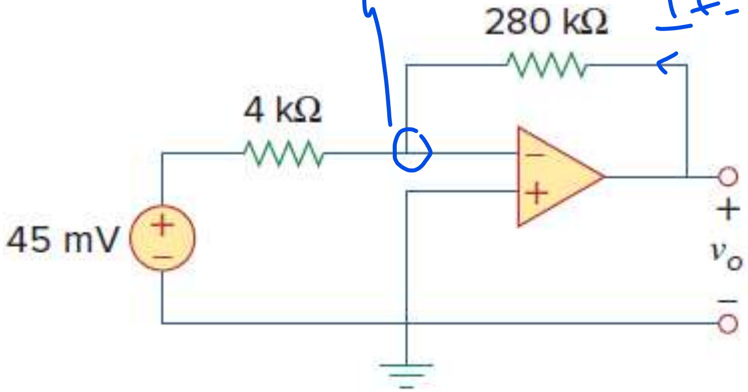
1)Refer to the op amp in Fig. below. If $v_i = 0.5$ V, calculate: (a) the output voltage v_o, and (b) the current in the 10-kΩ resistor.



2)Find the output of the op amp circuit shown in Fig. below. Calculate the current through the feedback resistor.

$V = 0$

$I_f = ?$



$$V_o = \left(\frac{-R_f}{R_{in}}\right) V_{in}$$

$$= \left(\frac{-\cancel{280}^{\,70}}{\cancel{4}}\right) 45$$

$$\boxed{V_o = 3.15}$$

$$I_f = \frac{3.15}{280} = 0.011$$

- Q3) Design a circuit with one op-amp that provides a gain of 5.5. Assume you have a resistor R_i =10kΩ, what value would you choose for R_f?

Q 4) Determine the input impedance and output voltage for the circuit in Figure

$$V_o = \left(-\frac{20}{5}\right) 100$$

$$4V$$

$$= +$$